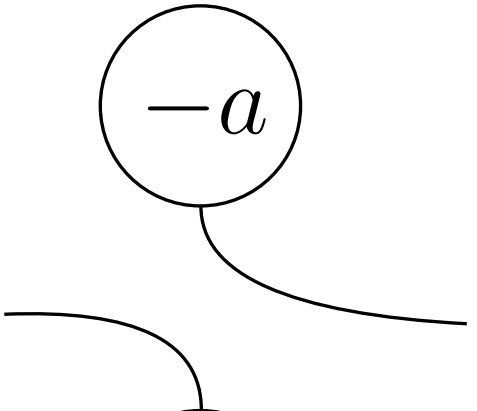

with $0 < a < b$